Normalization Practice Set

1. Given a relation R(A, B, C, D) and Functional Dependency set FD = { AB → CD, B → C }, determine whether the given R is in 2NF? If not convert it into 2 NF.

2. Given a relation R(P, Q, R, S, T) and Functional Dependency set FD = { PQ → R, S → T }, determine whether the given R is in 2NF? If not convert it into 2 NF.

3. Consider the following four relational schemas. For each schema, all non-trivial functional dependencies are listed, The underlined attributes are the respective primary keys.

   Schema I: Registration(rollno, courses) Field 'courses' is a set-valued attribute containing the set of courses a student has registered for. Non-trivial functional dependency rollno → courses
   Schema II: Registration (rollno, coursid, email) Non-trivial functional dependencies: rollno, courseid → email email → rollno
   Schema III: Registration (rollno, courseid, marks, grade) Non-trivial functional dependencies: rollno, courseid, → marks, grade marks → grade
   Schema IV: Registration (rollno, courseid, credit) Non-trivial functional dependencies: rollno, courseid → credit courseid → credit
   Which one of the relational schemas above is in 3NF

4. The relation scheme Student Performance (name, courseNo, rollNo, grade) has the following functional dependencies:

name, courseNo → grade
rollNo, courseNo → grade
name → rollNo
rollNo → name
Identify The highest normal form of this relation scheme. [Gate 2004]

5. Given a relation R(P, Q, R, S, T) and Functional Dependency set FD = { PQ → R, S → T }, determine whether the given R is in 2NF? If not convert it into 2 NF.

Given a relation R(P, Q, R, S, T, U, V, W, X, Y) and Functional Dependency set FD = { PQ → R, PS → VW, QS → TU, P → X, W → Y }, determine whether the given R is in 2NF? If not convert it into 2 NF.

6. Consider the following relation:

CAR_SALE(Car#, Date_sold, Salesperson#, Commission%, Discount_amt)
Assume that a car may be sold by multiple salespeople, and hence {Car#, Salesperson#} is the primary key. Additional dependencies are
Date_sold → Discount_amt and Salesperson# → Commission%
Based on the given primary key, is this relation in 1NF, 2NF, or 3NF? Why or why not? How would you successively normalize it completely?

7. Consider the following four relational schemas. For each schema, all non-trivial functional dependencies are listed, the underlined attributes are the respective primary keys.

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Schema IV: Registration (rollno, courseid, credit) Non-trivial functional dependencies: rollno, courseid → credit courseid → credit
Which one of the relational schemas above is in 3NF